

Overview of Bias variance tradeoff

Subject: Bias variance tradeoff

$$\text{MSE} = \text{Bias}^2 + \text{Variance} + \sigma_\epsilon^2$$

1.

Suppose that we have a training set consisting of a set of points $x_1, \dots, x_n$ and real-valued labels y_i associated with the points x_i . We assume that the data is generated by a function $f(x)$ such as $y = f(x) + \epsilon$, where the noise, ϵ , has zero mean and unit variance σ^2 . That is, $y_i = f(x_i) + \epsilon_i$, where ϵ_i is a noise sample. We want to find a function $\hat{f}(x; D)$, that approximates the true function $f(x)$ as well as possible, by means of some learning algorithm based on a training dataset (sample) $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$. We make "as well as possible" precise by measuring the mean squared error between y and $\hat{f}(x; D)$: we want $(y - \hat{f}(x; D))^2$ to be minimal, both for $x_1, \dots, x_n$ and for points outside of our sample. Of course, we cannot hope to do so perfectly, since the y_i contain noise ϵ ; this means we must be prepared to accept an irreducible error in any function we come up with. Finding an $\hat{f}$ that generalizes to points outside of the training set can be done with any of the countless algorithms used for supervised learning. It turns out that whichever function $\hat{f}$ we select, we can decompose its expected error on an unseen sample x (i.e. conditional on x) as follows:

2.

Approaches Dimensionality reduction and feature selection can decrease variance by simplifying models. Similarly, a larger training set tends to decrease variance. Adding features (predictors) tends to decrease bias, at the expense of introducing additional variance. Learning algorithms typically have some tunable parameters that control bias and variance; for example,

3.

$$\text{MSE}(x) = (f(x) - \mathbb{E}[\hat{f}(x)])^2 + \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - \hat{f}(x))^2] + \sigma^2 = \text{Bias}^2[\hat{f}(x)] + \text{Var}[\hat{f}(x)] + \sigma^2$$
$$\begin{aligned} \text{MSE}(x) &= \mathbb{E}[(f(x) - \hat{f}(x))^2] \\ &= \mathbb{E}[(f(x) - \mathbb{E}[\hat{f}(x)] + \mathbb{E}[\hat{f}(x)] - \hat{f}(x))^2] \\ &= \mathbb{E}[(f(x) - \mathbb{E}[\hat{f}(x)])^2 + 2(f(x) - \mathbb{E}[\hat{f}(x)])(\mathbb{E}[\hat{f}(x)] - \hat{f}(x)) + (\mathbb{E}[\hat{f}(x)] - \hat{f}(x))^2] \\ &= \mathbb{E}[(f(x) - \mathbb{E}[\hat{f}(x)])^2] + 2\mathbb{E}[(f(x) - \mathbb{E}[\hat{f}(x)])(\mathbb{E}[\hat{f}(x)] - \hat{f}(x))] + \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - \hat{f}(x))^2] \\ &= \text{Bias}^2[\hat{f}(x)] + \text{Var}[\hat{f}(x)] + \sigma^2 \end{aligned}$$

4.

$$E[(f(x) - \hat{f}(x))\epsilon] = E[f(x) - \hat{f}(x)]E[\epsilon] \text{ since } \epsilon \text{ is independent from } x = 0 \text{ since } E[\epsilon] = 0$$

5.

In human learning While widely discussed in the context of machine learning, the bias–variance dilemma has been examined in the context of human cognition, most notably by Gerd Gigerenzer and co-workers in the context of learned heuristics. They have argued (see references below) that the human brain resolves the dilemma in the case of the typically sparse, poorly-characterized training-sets provided by experience by adopting high-bias/low variance heuristics. This reflects the fact that a zero-bias approach has poor generalizability to new situations, and also unreasonably presumes precise knowledge of the true state of the world. The resulting heuristics are relatively simple, but produce better inferences in a wider variety of situations. Geman et al. argue that the bias–variance dilemma implies that abilities such as generic object recognition cannot be learned from scratch, but require a certain degree of "hard wiring" that is later tuned by experience. This is because model-free approaches to inference require impractically large training sets if they are to avoid high variance.

6.

where $N_1(x), \dots, N_k(x)$ are the k nearest neighbors of x in the training set. The bias (first term) is a monotone rising function of k , while the variance (second term) drops off as k is increased. In fact, under "reasonable assumptions" the bias of the first-nearest neighbor (1-NN) estimator vanishes entirely as the size of the training set approaches infinity.

7.

$$E[(y - \hat{f}(x))^2 | X = x] = (f(x) - \frac{1}{k} \sum_{i=1}^k f(N_i(x)))^2 + \sigma^2/k + \sigma^2$$

8.

The bias–variance tradeoff is a central problem in supervised learning. Ideally, one wants to choose a model that both accurately captures the regularities in its training data, but also generalizes well to unseen data. Unfortunately, it is typically impossible to do both simultaneously. High-variance learning methods may be able to represent their training set well but are at risk of overfitting to noisy or unrepresentative training data. In contrast, algorithms with high bias typically produce simpler models that may fail to capture important regularities (i.e. underfit) in the data. It is an often made fallacy to assume that complex models must have high variance. High variance models are "complex" in some sense, but the reverse need not be true. In addition, one has to be careful how to define complexity. In particular, the number of parameters used to describe the model is a poor measure of complexity. The model $f_{a,b}(x) = a \sin(bx)$ has only two parameters (a, b) but it can interpolate any number of points by oscillating with a high enough frequency, resulting in both a high bias and high variance. An analogy can be made to the relationship between accuracy and precision. Accuracy is one way of quantifying bias and can intuitively be improved by selecting from only local information. Consequently, a sample will appear accurate (i.e.

have low bias) under the aforementioned selection conditions, but may result in underfitting. In other words, test data may not agree as closely with training data, which would indicate imprecision and therefore inflated variance. A graphical example would be a straight line fit to data exhibiting quadratic behavior overall. Precision is a description of variance and generally can only be improved by selecting information from a comparatively larger space. The option to select many data points over a broad sample space is the ideal condition for any analysis. However, intrinsic constraints (whether physical, theoretical, computational, etc.) will always play a limiting role. The limiting case where only a finite number of data points are selected over a broad sample space may result in improved precision and lower variance overall, but may also result in an overreliance on the training data (overfitting). This means that test data would also not agree as closely with the training data, but in this case the reason is inaccuracy or high bias. To borrow from the previous example, the graphical representation would appear as a high-order polynomial fit to the same data exhibiting quadratic behavior. Note that error in each case is measured the same way, but the reason ascribed to the error is different depending on the balance between bias and variance. To mitigate how much information is used from neighboring observations, a model can be smoothed via explicit regularization, such as shrinkage.